

8. Ans: C

Gravitational Field Strength,  $g = \frac{GM}{r^2} \Rightarrow g \propto \frac{M}{r^2}$

Let the quantities with subscript 'J' represent that of Jupiter while those with subscript 'E' represent that of Earth.

$$\begin{aligned}\therefore \frac{g_J}{g_E} &= \frac{M_J}{M_E} \left( \frac{r_E}{r_J} \right)^2 \\ \Rightarrow \frac{g_J}{9.81} &= \frac{314}{1} \left( \frac{6400 \times 10^3}{7.23 \times 10^7} \right)^2 \\ \therefore g_J &= 24.1 \text{ m s}^{-1}\end{aligned}$$